Pinpoint: an AR build guide for electronics and LEGO learners

Final Year Project overview · Nikol Staykova · 7 October 2026

The idea

Pinpoint turns an Arduino or Raspberry Pi tutorial or a LEGO booklet into a graph of the parts and how they connect. From that graph, a phone app uses augmented reality to create real-time, adaptive lessons: through the camera it shows on the real build where the next part goes, checks each connection as it is made, and adapts to how the learner builds. Pinpoint checks the result, not the order. Any correct way of building is accepted. To do this in the cheapest and most efficient way, it uses different algorithms and models throughout.

Target groups

- **Arduino and Raspberry Pi learners** (first market): students and hobbyists who get stuck on wiring.
- **STEM teachers:** a tutor at every lab bench.

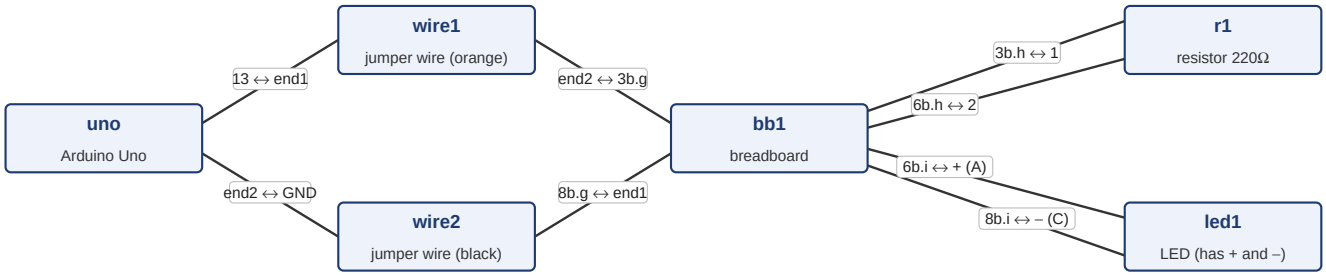
How the project developed

The project grew out of **CircuitQuest**, my app that checks Arduino wiring. The checking engine of CircuitQuest only needs parts, their pins and the connections between them, so the idea was: turn any instructions into that kind of graph and the same checking works for other builds too. I planned the work in steps:

- **1. Research** how instructions and assemblies are described and survey manuals (furniture, LEGO, Arduino).
- **2. Design the graph:** what the nodes, edges and rules are.
- **3. Build the graph generator**, then improve it version by version, measuring accuracy, time and cost.
- **4. Prepare the output for AR:** ordered steps, positions and live checks.

What the graph is

The Blink circuit as a graph: Arduino pin 13 → orange wire → resistor → LED → black wire → GND. Each line joins two named points ("3b.g" = breadboard column 3, bottom half, row g).



One graph per manual.

- Nodes are the physical parts: the board, each wire, resistor, LED, LEGO brick. A wire is a node with two ends, so it has two edges, one for each end.
- Edges are connections between named points on two parts (pin 13, hole 3b.g, a stud). They are typed: joined (held together) or electrical (current can flow); a wire in a breadboard has both.
- Edges are mostly undirected; direction is added only where it matters, like an LED's + and – legs.
- A build is correct if it makes the same connections, so a different but equivalent hole or order is fine.
- Each node has a name (r1, led1, b12), a type from a part catalogue (resistor, LED, LEGO brick 2x4) and its own properties, such as colour or resistor value. Every physical item counts, down to each wire and pin.
- The part catalogue says what each type is like inside: its named connection points, which points are electrically the same (the 5 holes of a breadboard strip, a wire's two ends), which are interchangeable (a resistor's legs) and which parts have a + and – (an LED).
- Each edge also records how the connection is made (pushed in, clicked onto studs, screwed), whether it can move (rigid or a hinge) and whether it can be undone by hand.
- From the graph, code computes the electrical circuits and the solid blocks that hold together. A build is judged on these, which is why a resistor in a different but connected hole still counts as right, while an LED put in backwards is always wrong.
- Simple rules check the graph itself: the part counts match the parts list, every part is used, and each connection point exists on its part and is used only as often as it can be.

How the graph generator works

- It begins with the **parts list**. Then it creates the connections between them.
- For LEGO sets that have a 3D model, the graph is built directly from the model using code.
- For sets without a model, **Claude reads the instructions**. Then code checks the result against physical and electrical rules and sends back any errors that need fixing.

Results so far

- **Electronics:** it works very well. The best version had no real errors on the tutorials we reviewed, and the output is ready for use in AR.
- **LEGO:** the output is exact when a 3D model is available. When Claude has to read the booklet, the connections are usually correct, but the exact positions are not yet reliable enough for AR. That is what I am working on improving now.
- Time and cost are measured for every run, along with accuracy, to track performance.